Dimensional Aspect Sentiment Regression Using a BERT-Based Approach

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Abstract

Traditional aspect-based sentiment analysis typically maps textual features to discrete polarity categories. This research studies Dimensional Aspect Sentiment Regression (DimASR), a SemEval task that requires predicting continuous, real-valued scores for Valence (pleasantness) and Arousal (intensity) for particular aspects in review texts. A comparative assessment is performed using the official English-language dataset, contrasting a Support Vector Regressor (SVR) baseline with a fine-tuned Transformer architecture (BERT). By implementing Huber Loss to reduce Gaussian distribution bias, applying strict coordinate constraints, and applying early stopping to prevent capacity mismatch, the contextual BERT model obtains a Root Mean Square Error (RMSE) of 0.8912, considerably outperforming the lexical baseline. A detailed metric interpretation justifies sub-1.0 error limits, and an error analysis assesses the connection among aspects.

1 Introduction

Aspect-based sentiment classification conventionally seeks to detect sentiment polarity toward specific entity features (Pontiki et al., 2014). For example, given the review "The camera quality is amazing, but the battery life is terrible," a standard model assigns a positive label to the camera and a negative label to the battery. However, human emotion is better represented as a spectrum rather than as discrete categorical boundaries.

The Dimensional Aspect Sentiment Regression (DimASR) task models emotion using a two-dimensional Valence-Arousal (VA) spatial representation (Russell, 1980). In this system, Valence quantifies pleasantness, and Arousal quantifies intensity. As a result, emotions are represented as coordinates: excitement corresponds to high valence and high arousal, while calmness indicates high valence but low arousal (Russell, 2003).

This project paper documents the procedure for the DimASR regression task. Given an input sentence and a target aspect, the system predicts continuous VA scores ranging from 1.00 (extremely negative/low) to 9.00 (extremely positive/high), with 5.00 indicating neutrality. Two predictive architectures are evaluated: a term-frequency baseline and a deep bidirectional Transformer. The evaluation focuses on their capacity to distinguish multi-aspect sentiments and accurately minimize the Root Mean Square Error (RMSE) within the continuous emotional space.

2 Related Work

Early research in Aspect-Based Sentiment Analysis (ABSA) relied heavily on lexicon matching and syntactic dependency parsing to extract and classify sentiment (Pontiki et al., 2016). Recent surveys indicate an almost complete transition to deep neural networks and pre-trained language models, which more effectively handle implicit aspects and multi-clause sentiment routing (Zhang et al., 2023).

Concurrently, dimensional emotion analysis has earned attention for its psychological validity. Resources such as EmoBank (Buechel and Hahn, 2017) and continuous emotion tasks in WASSA (Preoţiuc-Pietro et al., 2016; Mohammad and Bravo-Marquez, 2017) demonstrate that mapping text into real-valued emotional intensities provides greater analytical granularity than binary classification. This research connects these domains by applying contextual embeddings to regress aspect-specific coordinates in the continuous VA space.

3 Dataset and Task Formulation

3.1 Data Filtering and Preprocessing

The data for this task originates from a multilingual, multi-subtask release. In accordance with task constraints, a rigorous post-processing pipeline was implemented. First, English-language instances were isolated. Second, the dataset was filtered to retain only data relevant to Subtask 1 (DimASR), excluding quadruple-extraction tasks.

This process resulted in a working corpus of 2,779 training samples, 340 validation samples, and 1,504 test samples.

3.2 Input and Output Parsing

The official task format uses JSON Lines with a well-defined format. Inputs include an `ID`, the review `Text`, and a list of target `Aspect` strings. The required system output must preserve the `ID` and the exact aspect casing and order, appending an `Aspect_VA` list containing predicted coordinates formatted as a `V#A` string (e.g., "6.75#6.38"), strictly rounded to two decimal places. A custom PyTorch Dataset class was developed to parse this structure, converting string targets into floating-point tensors during training and formatting them back to the exact task specifications during inference. A neutral 5.0#5.0 fallback was applied to safely process unannotated variables during blind testing.

4 Methodology

4.1 Evaluation Metric

In continuous regression tasks, performance is measured exclusively using Root Mean Square Error (RMSE). This metric heavily penalizes severe misclassifications, which is essential during modeling high-intensity emotional outliers. The error is calculated simultaneously across the two-dimensional emotional space:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}((V_p - V_g)^2 + (A_p - A_g)^2)} \quad (1)$$

where p represents the predicted score and g represents the gold standard annotation.

4.2 Baseline SVR Model

To establish a lexical lower bound, a Support Vector Regressor (SVR) was deployed. The input representation consists of TF-IDF vectors (maximum 3,000 features, 1-2 n-grams) derived from concatenating the review text and the target aspect. Because SVR is inherently single-output, two independent regressors were trained: one for Valence and one for Arousal.

4.3 Contextual Transformer Model

To capture syntactic nuance, the `BERT-base-cased` architecture was fine-tuned. The cased variant was explicitly selected because capitalization frequently influences Arousal intensity (e.g., "GREAT" vs. "great"). The standard sequence classification head was replaced with a linear layer that outputs two continuous values. For multi-aspect routing, inputs were formatted as `[CLS] Text [SEP] Aspect [SEP]`, prompting the self-attention matrices to compute target-specific context.

5 Results and Comparative Analysis

Table 1 presents the empirical performance of both models on the development set.

Model	Feature Representation	Validation RMSE
SVR	TF-IDF (Lexical)	1.1465
BERT-base	Transformer	**0.8912**

Table 1: Validation RMSE Performance

5.1 Metric Interpretation and Significance

When evaluated in isolation, an RMSE approaching 1.0 may appear elevated compared to metrics in discrete classification tasks. However, contextualizing this error margin demonstrates strong forecasting precision. The DimASR task uses an 8-point continuous scale (1.00-9.00). An RMSE of 0.8912 signifies an average deviation of approximately 11% from the true emotional coordinate.

Furthermore, continuous emotion annotation is essentially subjective. Human annotators rarely assign identical real-valued intensities to affective modifiers, resulting in a natural floor in the ground-truth label variance. Achieving a sub-1.0 RMSE indicates that the architecture reliably captures true linguistic magnitude and likely approaches the noise floor of inter-annotator disagreement.

The primary indicator of success is the relative performance delta. The transition from the lexical SVR baseline (1.1465) to the contextual Transformer (0.8912) yields an approximate 22% reduction in error. This result quantitatively confirms that deep contextual representations resolve complexities missed by term-frequency overlaps.

5.2 Error Analysis

Although the overall RMSE is highly competitive, a qualitative analysis of the validation and test predictions reveals specific behavioral trends:

- **Magnitude Overestimation on Positive Aspects:** While the model reliably maps polarity, it at times struggles with exact magnitude. For instance, in the validation set, it correctly identified positive sentiment for "diner food" and "breakfast", but overestimated both Valence and Arousal (predicting $\sim$7.70 against a gold standard of 7.25#6.75).

- **Lingering Neutrality Pull in Multi-Aspects:** Despite the implementation of Huber Loss, there is a slight lingering bias toward the dataset mean. In the blind test set for a review describing a pizza as "pretty bland", the model efficiently depressed the Valence scores for "pepper pizza", "pepperoni", and "sausage" (4.25 to 4.43), but clustered the Arousal scores tightly around 6.05, indicating some hesitation to predict extremely low-arousal coordinates.

6 Analysis of Model Challenges

During development, several mathematical and structural challenges were identified and addressed:

1. Aspect Cross-Talk: In multi-clause reviews, standard models conflated sentiment across individual entities. The `[SEP]`-delineated aspect token in the Transformer architecture localized attention parameters to the relevant modifiers in the sequence, thereby isolating the emotion.

2. Regression to the Mean: The distribution of the English restaurant dataset is heavily Gaussian, clustered around the neutral 5.0 score. When the standard Mean Squared Error (MSE) loss was used, the neural network learned to conservatively predict values near the mean to minimize average penalties, thereby ignoring high-arousal outliers. This issue was mitigated by replacing MSE with Huber Loss (`SmoothL1Loss`), which is more robust to outliers and enables the model to map extreme coordinates with greater confidence.

3. Capacity Mismatch and Overfitting: Fine-tuning a 110-million-parameter BERT model on a restricted training corpus of 2,779 samples presented a substantial risk of capacity mismatch and overfitting. To stabilize convergence, strict regularization was implemented, including increased weight decay (0.1) with the AdamW optimizer and early stopping based on validation loss plateaus. The model reached optimal generalization by Epoch 3.

4. Output Range Violations: Neural regressors generate unbounded real numbers $(-\infty, +\infty)$. However, valid task coordinates must strictly fall within $[1.00, 9.00]$. This was addressed by applying a bounding constraint function immediately post-inference, using a NumPy clipping array to force all extreme values to the valid boundaries before final two-decimal string formatting.

7 Conclusion

This project paper validates the efficacy of deep contextual architectures over lexical baselines for dimensional sentiment regression. By filtering the multilingual corpus, enforcing spatial limits, and using Huber Loss to address dataset distribution biases, the fine-tuned BERT model reached an RMSE of 0.8912, effectively matching human-level variance on a continuous scale.

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A Core Implementation Logic

A.1 Coordinate Bounding and Formatting

```
import numpy as np

# post inference boundary enforcement
test_preds = np.clip(test_outputs.predictions, 1.0, 9.0)

# formatting for V#A string (to 2 decimal places)
formatted_va = []
for v, a in test_preds:
    va_str = f"{v:.2f}#{a:.2f}"
    formatted_va.append(va_str)
```

A.2 Huber Loss Integration

```
import torch.nn as nn
from transformers import Trainer

class DimASRTrainer(Trainer):
    def compute_loss(self, model, inputs,
                return_outputs=False, **kwargs):
        labels = inputs.pop("labels")
        outputs = model(**inputs)
        logits = outputs.get("logits")

      # Huber Loss to mitigate Gaussian bias
        loss_fct = nn.SmoothL1Loss()
        loss = loss_fct(
        logits.view(-1, self.model.config.num_labels),
        labels.view(-1, self.model.config.num_labels)
        )
      return (loss, outputs) if return_outputs else loss
```